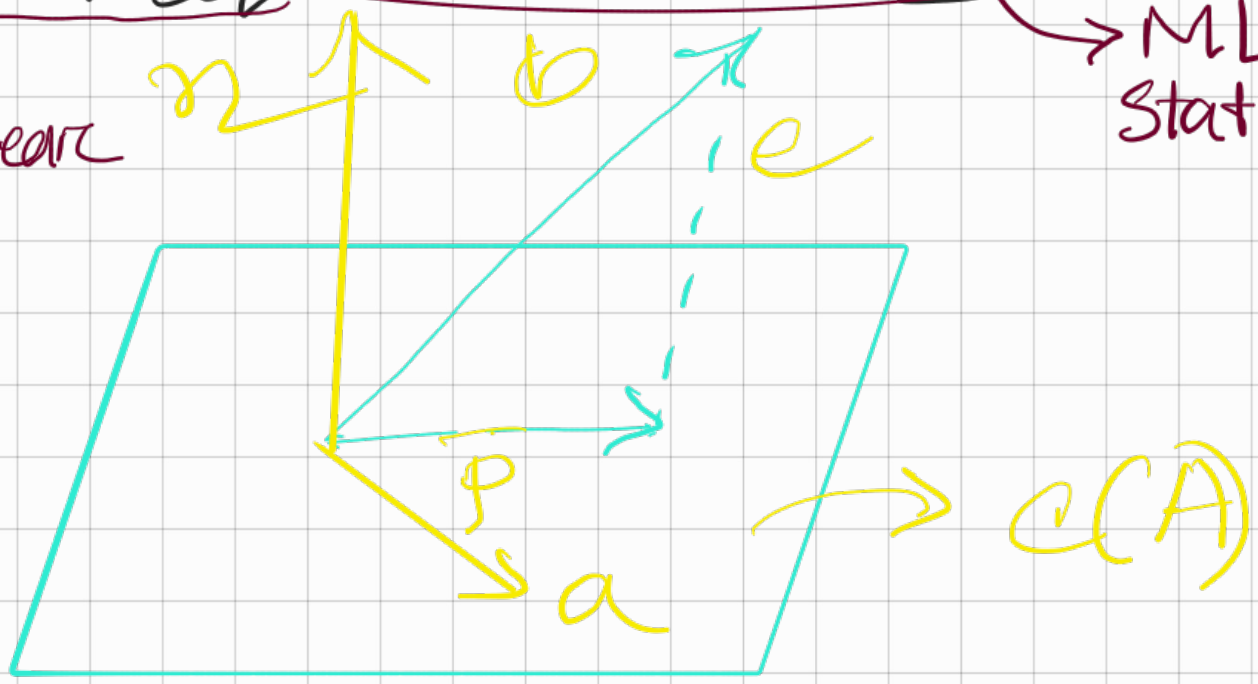
Least sq regression → ML Statistics

or Linear



$Pa = a$ ✓ → $N(A^T)$

$Px \rightarrow$ Perpendicular to $C(A)$

$Px = 0$

$Ax = 0$

$Pb = p$

$Ax = b$

$A^T A\hat{x} = A^T b$ ✓

$$P \cdot A(A^TA)^{-1}A^T x = 0$$

$$b = p + e$$

$$\Rightarrow e = b - p$$

$$e = b - Pb$$

powerful

$$e = b(I - P)$$

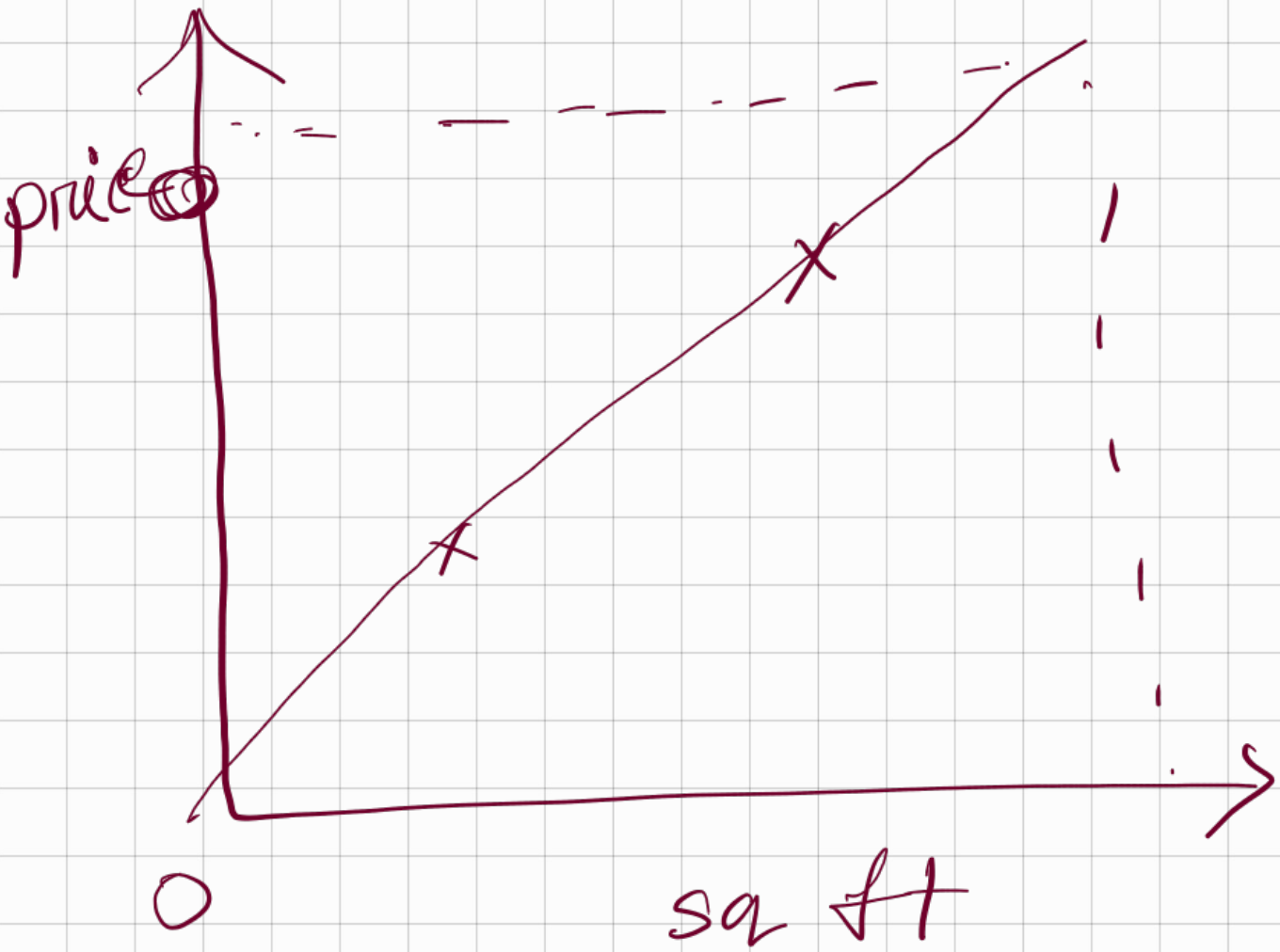
↳ projects onto left nullspace

* Find a vector that is perpendicular to the

col space of Mat A

↳ Qstn.

Transforming b into perpendicular vector e.



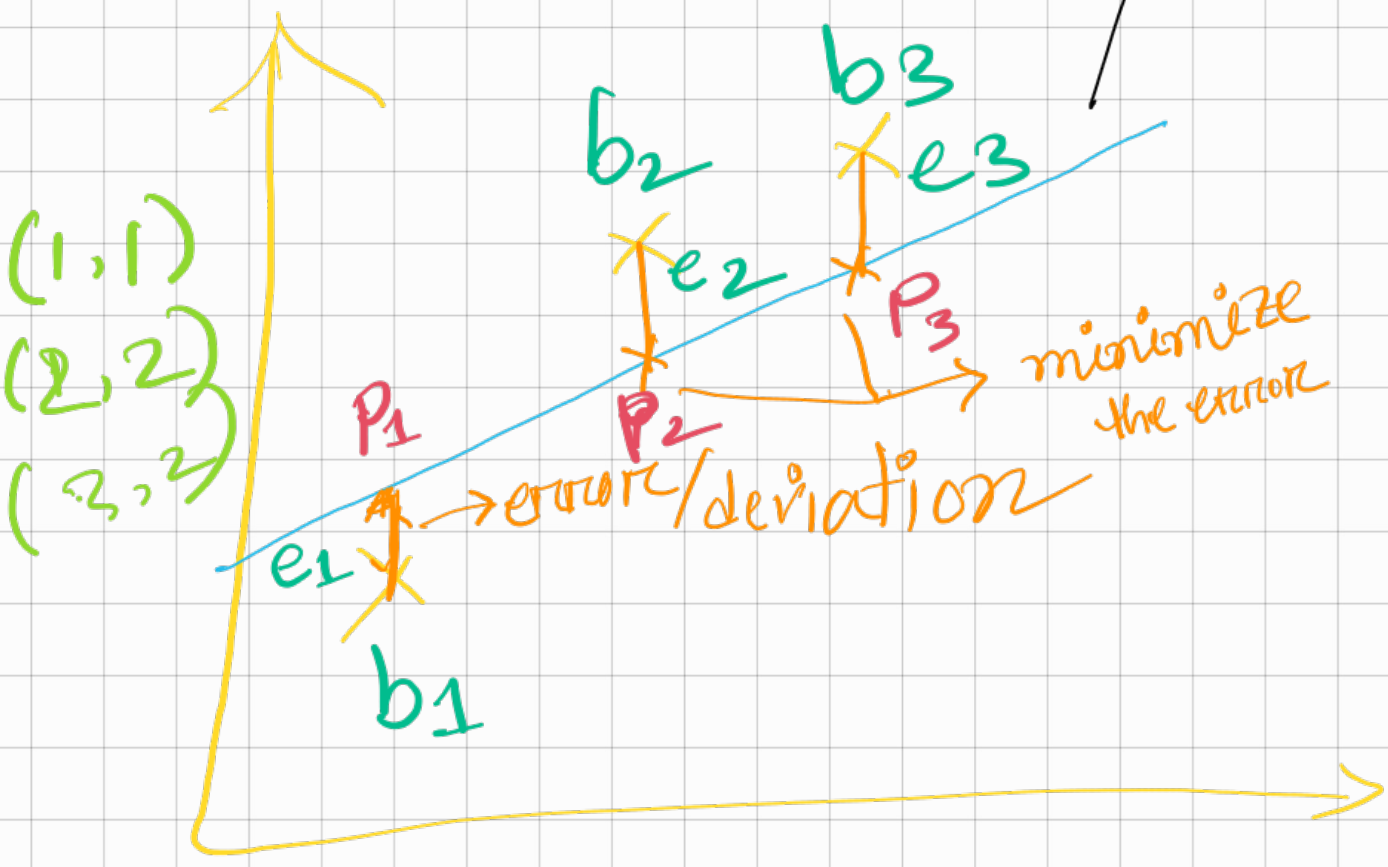
2 data points

ML → Feeding data then predicting what's gonna come next.

For 2 points

$y = Dx + C$

$y = \frac{1}{2}x + \frac{2}{3}$



Some Metrics that will tell whether a line is

good or bad.

So minimize the length of the error vectors.

$$C+D = 1$$
$$C+2D = 2$$
$$C+3D = 2$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

A $\quad$ x $\quad$ b

↓

no x for which Ax = b true

$y = mx + c$

for (1, 1)

$1 = m + c$

minimize total error

$e_1 + e_2 + e_3$

↓ ↓ ↓

$-3 \quad 2 \quad 1$

sum = 0 but error not 0.

So modulus or better square root but best outliers. to get rid of the sign.

$e_1^2 + e_2^2 + e_3^2$ → least of the square

↓

→ to fit a linear line.

minimize it

whole would individually minimize

$$A^T A \hat{x} = A^T b$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} & \end{bmatrix}$$

$A^T A$

symmetric across diagonal.

→ Invertible otherwise can't get $\hat{x}$.

$1+1+1 \quad 1+2+3$

$\begin{bmatrix} 3 & 6 \end{bmatrix}$

$1+2+3 \quad 1+4+6$

$\begin{bmatrix} 6 & 14 \end{bmatrix}$

$42-36=6$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$A^T b$

$1+2+2$

$1+4+6$

We get $\Rightarrow$ $\rightarrow \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix}$

$$\therefore \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} \hat{x} \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$D = \frac{1}{2}$

$D = \frac{1}{2}$ ✓

$3C + 3 = 5$

$C = \frac{2}{3}$ ✓

$$\boxed{\begin{aligned} C &= \tfrac{2}{3} \\ D &= \tfrac{1}{2} \end{aligned}} \longrightarrow \hat{x}$$

$\therefore y = \frac{1}{2}x + \frac{2}{3}$ → best fit line.

$e_1^2 + e_2^2 + e_3^2$

$= (b_1 - P_1)^2 + (b_2 - P_2)^2 + (b_3 - P_3)^2$ ✓

local minima & maxima → H.W

for the error func also gives the best fit line.

Non linear regression.



If A has independent columns, then A^TA is invertible. ✓

$$A^TA\hat{x} = 0$$

↓

if A has a x other than 0 then col are dependable

if for $A^TAx = 0$, x is only 0 we can say A^TA is invertible.

$A^T A x = 0$

↓

happens 1st

So x must be 0